Chapter 2: Sequences

Intro: Recall in the early lectures we claimed that infinite series, like

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \cdots$$

were useful for solving a variety of problems. Other series could be consisting of numbers such as

$$1 + 1/2 + 1/4 + 1/8 + \cdots$$

In this case, this series gets closer and closer to 2. So, it is sometimes written that

$$1 + 1/2 + 1/4 + 1/8 + \cdots = 2.$$

The way to think about this is as a sequence given by

$$a_1 = 1$$

$$a_2 = 1 + 1/2 = 3/2$$

$$a_3 = 1 + 1/2 + 1/4 = 7/4$$

At each step, we define a term in a sequence. In this case the sequence is given by

$$\left(1, 3/2, 7/4, 15/8, 31/16, \ldots\right)$$

2-Sequences

A sequence is simply an ordered list of numbers, with no end. Ex:

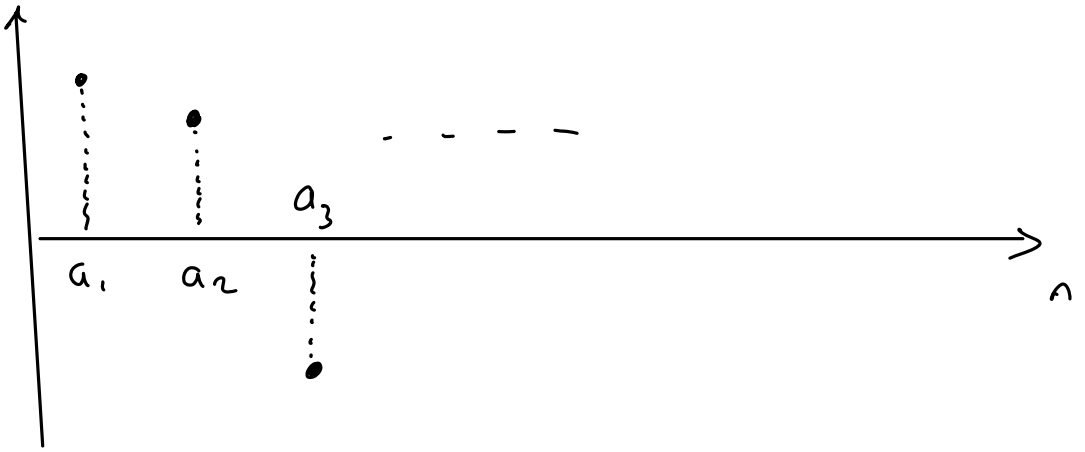
$$(1, 4, 9, 16, \ldots)$$

$$(\pi, 3, \sqrt{2}, e, 10, \ldots)$$

Defn 2.1 : Sequence

A **sequence** is a function $a : \mathbb{N} \to \mathbb{R}$.

Pic :



Remark 2.2 :

i) Since $a: \mathbb{N} \to \mathbb{R}$, we would normally write something like $a(n)$. Instead we write it with subscripts, that is a_n. This is known as the n-th term of the sequence. We write the individual terms as

$$a_1 = 1,\ a_2 = 4,\ a_3 = 9,\ a_4 = 16, \ldots$$

ii) In some special cases, the terms of a given sequence can be written explicity. For example, the above can be written as

$$a_n = n^2.$$

iii) Some sequences can be defined recursively. For example the sequence

$$a_n = a_{n-1} + a_{n-2}$$

with $a_1 = a_2 = 1$. This defines the Fibonacci sequence, and is given by

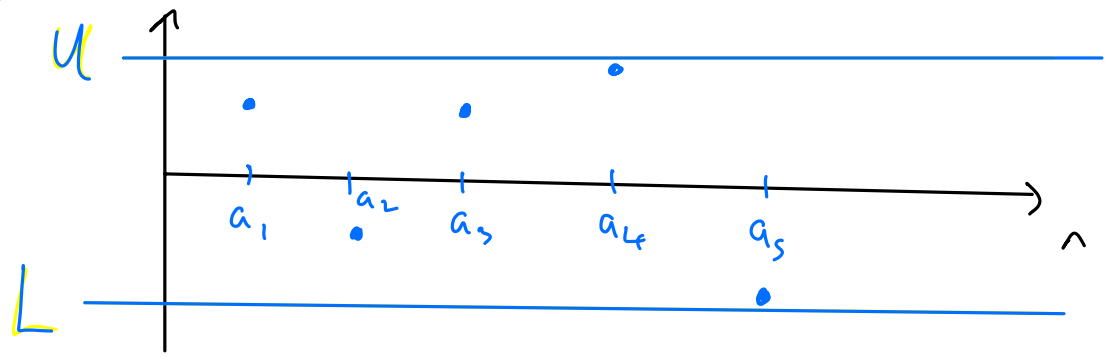
$$(1, 1, 2, 3, 5, \ldots)$$

iv) In general, a sequence will be written as (a_n). Sometimes it can be written as

$(a_1, a_2, a_3, \ldots)$ or $(a_n)_{n=1}^{\infty}$.

3- Bounded Sequence

If I think about this in terms of the definition of bounded, I would that a bounded sequence is one that is constrained to some region.



Defn 3.1 : Bounded Sequence

Let (a_n) be a sequence. Then (a_n) is bounded above if there exists $U \in \mathbb{R}$ such that $a_n \leq U$ for all $n \in \mathbb{N}$.

Similarly for bounded below (See notes).

Finally, a sequence (a_n) is bounded if it is bounded from both above and below. If a sequence is not bounded, we say it is unbounded.

Theorem 3.2 : Bounded Sequence

A sequence (a_n) is bounded iff there exists $K \in \mathbb{R}$ s.t. $|a_n| \leq K \quad \forall n \in \mathbb{N}$.

Proof: Exercise.

Ex 3.3 : (ii) Let (b_n) be a sequence given by $b_n = 1 + \sin(n)$. Then (b_n) is bounded.

Proof: Since, $|\sin(n)| \leq 1 \quad \forall n \in \mathbb{N}$ it follows that

$$|b_n| = |1 + \sin(n)| \leq |1| + |\sin(n)|$$
$$\leq 1 + 1$$
$$= 2.$$

Hence, (b_n) is bounded. ☑

(iii) Let (c_n) be the sequence given by $c_n = (-1)^n n$.

Then (c_n) is unbounded.

Proof: Assume towards a contradiction that (c_n) is bounded. That is, there exists $k \in \mathbb{R}$ s.t

$$|c_n| = |(-1)^n n| \leq k$$

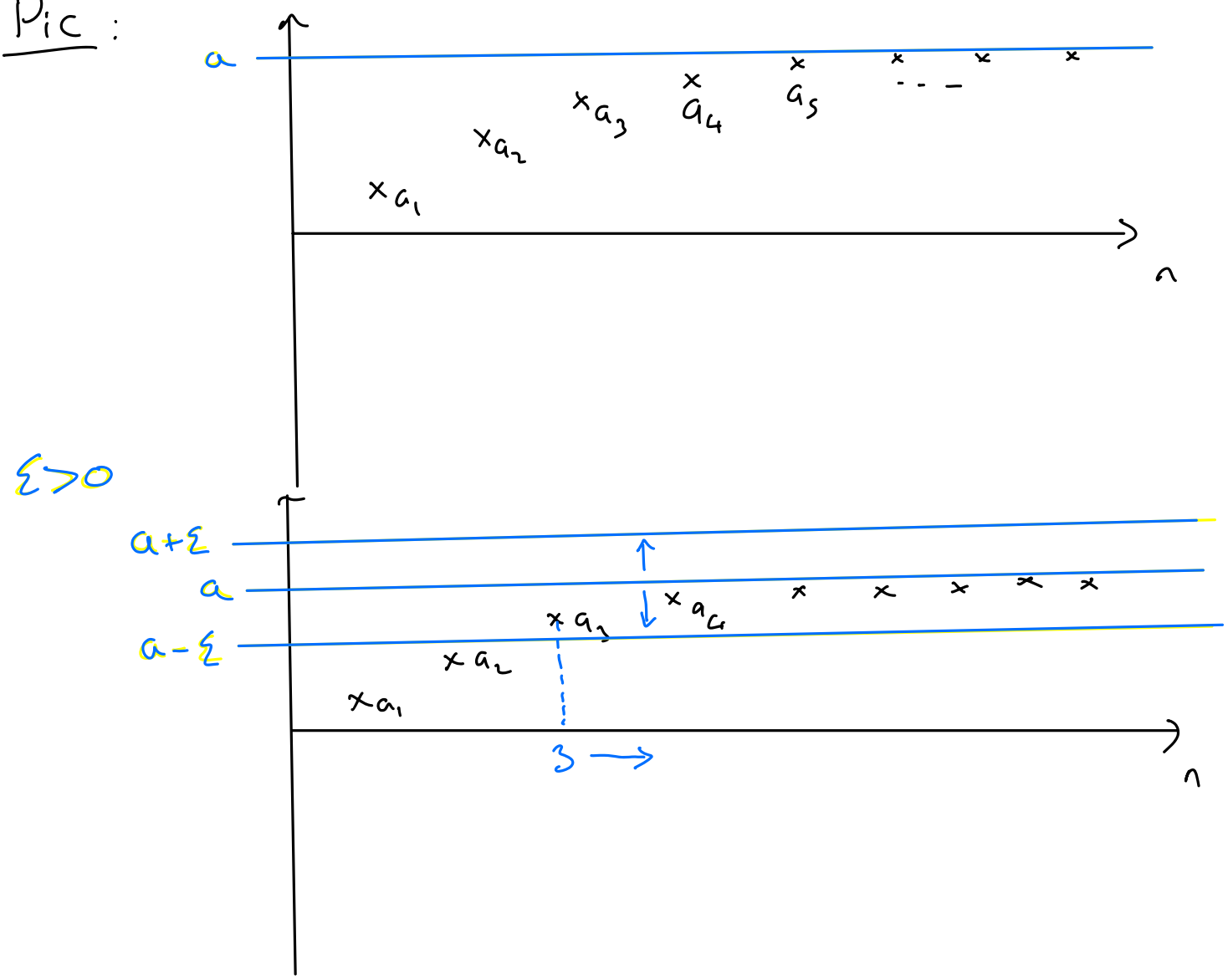
for all $n \in \mathbb{N}$. By properties of the absolute value function, we have that

$$|c_n| = n \leq k, \text{ for all } n \in \mathbb{N}.$$

But this contradicts the Archimedean property. Hence, (c_n) is unbounded. ☐.

Convergent Sequences

Pic:



Defn 4.1: ε-Neighbourhood

Let $\varepsilon > 0$. Then the ε-neighbourhood $V_\varepsilon(a)$ of a point $a \in \mathbb{R}$ is the interval

$$V_\varepsilon(a) := (a-\varepsilon, a+\varepsilon) = \{x \in \mathbb{R} : a-\varepsilon < x < a+\varepsilon\}.$$

Defn 4.2 : (Topological Convergence)

A sequence (a_n) converges to $a \in \mathbb{R}$ if, given any ε-neighbourhood of a, there exists a point in the sequence after which all of the terms are in $V_\varepsilon(a)$.

Defn 4.3 : (Metric Convergence)

A sequence (a_n) converges to $a \in \mathbb{R}$ if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon$$

for all $n > N$.

Remark : Suppose (a_n) converges to a. We can write this as

i) $\lim(a_n) = a$

ii) $a_n \to a$ as $n \to \infty$

iii) $a_n \to a$

iv) $\lim_{n\to\infty}(a_n) = a$.

Ex: Let (a_n) be the sequence given by $a_n = 1/n$.

Then $\lim(a_n) = 0$.

Proof: By the Archimedean property, it follows that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t.

$$1/N < \varepsilon.$$

Hence, for every $n > N$ it follows that

$a = \lim(a_n) = 0$.

$$|a_n - 0| = |1/n|$$

$$< \frac{1}{N}$$

$$< \varepsilon.$$

Thus $\lim(a_n) = 0$. ☐.

Ex: Let (a_n) be the sequence given by $a_n = \frac{3n+1}{n+2}$. Then $\lim(a_n) = 3$.

Scratch Work:

Want to construct an inequality of the form

$$|a_n - a| < \frac{K}{n}$$

for some $K \in \mathbb{R}$. For this example

$$|a_n - a| = \left| \frac{3n+1}{n+2} - 3 \right|$$

$$= \left| \frac{3n+1 - 3(n+2)}{n+2} \right|$$

$$= \left| \frac{-5}{n+2} \right|$$

$$= \frac{5}{n+2}$$

$$< \frac{5}{n}.$$

Proof: By the Archimedean property, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$5/N < \varepsilon .$$

Hence, $\forall n > N$ we have that

$$|a_n - 3| < 5/n < 5/N < \varepsilon .$$

Ex: $a_n = \dfrac{3n^2 + 2n - 10}{n^2 - 3n - 3}$. $\lim(a_n) = 3$.

Scratch Work:

$$|a_n - a| = \left| \frac{3n^2 + 2n - 10}{n^2 - 3n - 3} - 3 \right|$$

$$= \left| \frac{3n^2 + 2n - 10 - 3(n^2 - 3n - 3)}{n^2 - 3n - 3} \right|$$

$$= \left| \frac{11n - 1}{n^2 - 3n - 3} \right|$$

$$= \frac{11n - 1}{|n^2 - 3n - 3|}$$

$$< \frac{11n}{|n^2 - 3n - 3|}.$$

We wish to find some $N \in \mathbb{N}$ such that

$$\tfrac{1}{2} n^2 < n^2 - 3n - 3$$

for all $n \geq N$. This can be rearranged to give

$$0 < n^2 - 6n - 6.$$

This is true for all $n \geq 7$. Hence, if $n \geq 7$, $|a_n - 3| < \frac{11n}{n^2/2} = \frac{22}{n}$.

Proof: By the A.P., given $\varepsilon > 0$, there exists $N_1 \in \mathbb{N}$ s.t

$$\frac{22}{N_1} < \varepsilon.$$

Let $N = \max\left(7, \frac{22}{N_1}\right)$. Then it follows

$\forall n > N$ we have that

$$|a_n - 3| < \frac{22}{n} < \frac{22}{N} < \varepsilon . \blacksquare$$